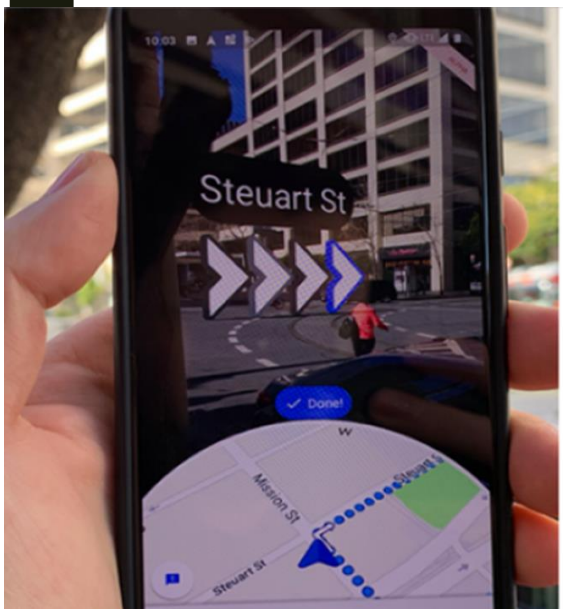
THE INTERACTION

Chapter 4



The Interaction

- **Interaction models**
  - *translations between user and system*
- Ergonomics
  - *Physical characteristics of interaction*
- Interaction styles
  - *the nature of user/system dialog*
- Context of interaction
  - *social, organizational, motivational*
- Paradigms
  - *interactive system design paradigms for usable designs*

What is interaction?

- Interaction is the communication between the user and system.

User ⟷ System

- Interaction is said to be occur when:
  - *the human user uses the computer as a tool to perform, simplify or support a task.*
  - *a user communicate his/her requirements to the computer*
  - *a user constantly providing instruction (to the computer/system) and receiving feedback (from the computer/system)*

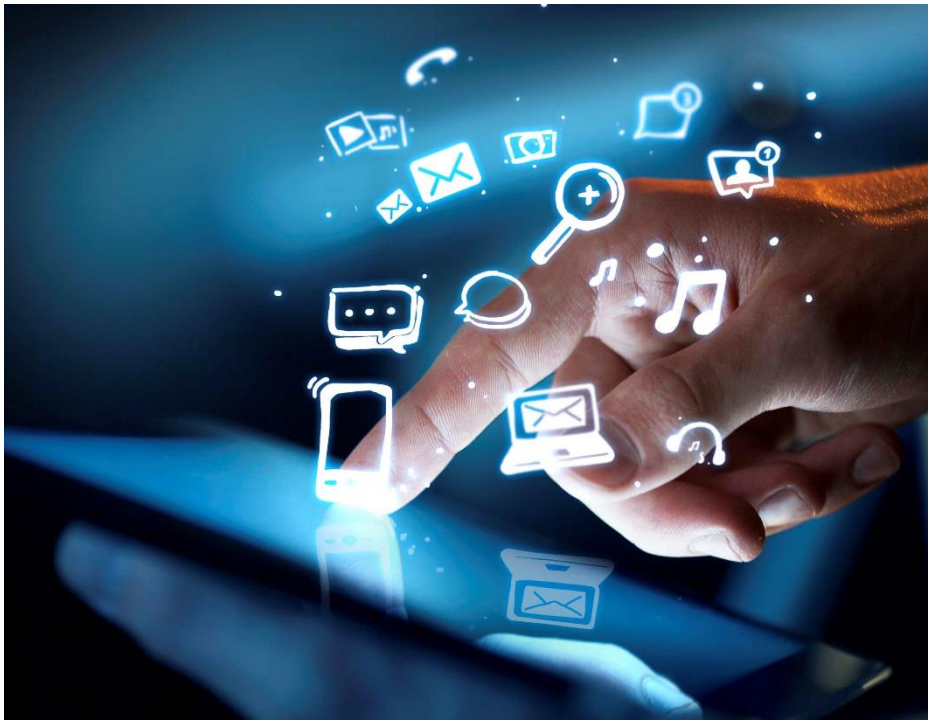
    e.g. … select fill tool, click over triangle

Models of interaction

- Interaction models shows an abstract view of what is going on in the interaction between user and system.
  - *address the translations between what the user wants and what the system does*
- Terms of interaction
  - ***Domain*** *– the area of work under study e.g. graphic design*
  - ***Goal*** *– what you want to achieve e.g. create a solid red triangle*
  - ***Task*** *– how you go about doing it? operations or actions to achieve a goal*
    - *e.g. … select fill tool,, drawing a triangle*
  - ***Intention*** *- intention is a specific action required to meet the goal*

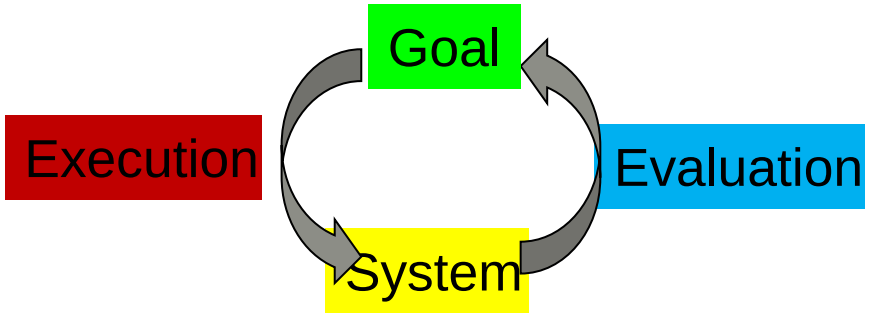
Models of interaction

- The purpose of an interactive system is to aid a user in accomplishing *goals from* some application *domain*
- Building an interaction model involves *task analysis* – the description of the problem space for the user of an interactive system in terms of the domain, goals, intentions and tasks.



Models of interaction....

- Donald Norman's model
  - *Seven stages of interaction*
    - User establishes the goal
    - Formulates intention
    - Specifies actions at interface
    - Executes action
    - Perceives system state
    - Interprets system state
    - Evaluates system state with respect to goal



Models of interaction....

- Execution/evaluation loop
  - *The user formulated plan/goal*
  - *The user's notion translated into more specific intention and actual actions*
  - *The formulated plan is executed by the computer.*
  - *The user perceives (finds) a resulting state*
  - *The user interprets the result against his expectations (goals)*
  - *the user evaluates the results (if goal, the computer has done) and determines the further actions.(new goals) Or repeat the cycle (if not goal)*

- *Norman's model concentrates on user's view of the interface only*

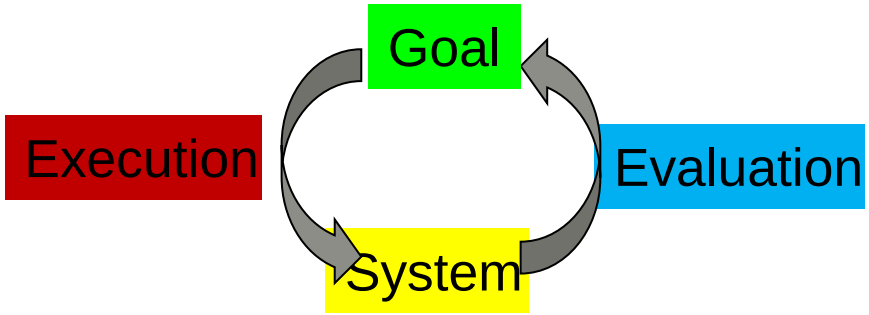


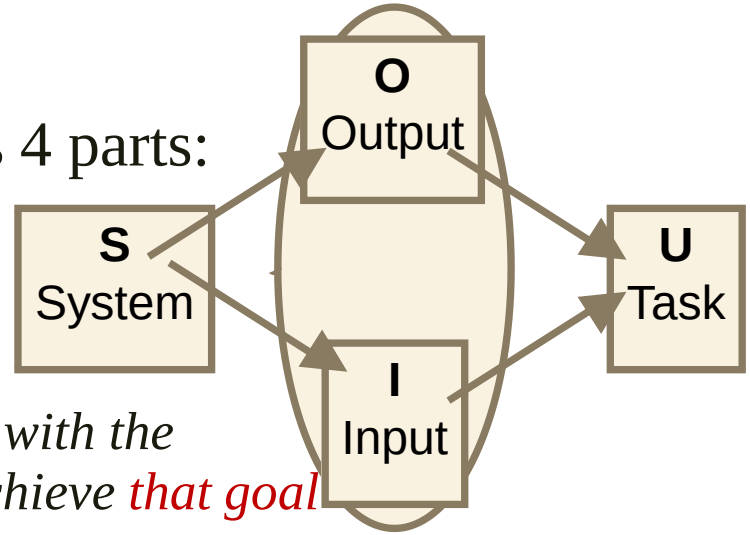
Illustration: switching on a light

- Imagine you are sitting reading as evening falls.
  - *You decide you **need more light**; that is you **establish the goal** to get more light.*
  - *From there you **form an intention** to **switch on the desk lamp**, and you **specify the actions required**, to **reach over and press the lamp switch**.*
  - *When you have **executed the action** you **perceive** the **result**, either the light is **on** or it **isn't;***
  - *Then you **interpret** this, based on your knowledge of the world. For example, if the light does not come on you may interpret this as indicating the bulb has blown or the lamp is not plugged into the mains, and you will formulate new goals to deal with this.*
  - *If the light does come on, you will evaluate the new state according to the original goals – is there now enough light? If so, the cycle is complete. If not, you may formulate a new intention to switch on the main roof light as well.*

Human error - slips and mistakes

- Created during the execution-evaluation loop
  - *Slip*
    - Understand system and goal
    - Correct formulation of action
    - Incorrect action
  - *Mistake*
    - May not even have right goal!
  - *Fixing things?*
    - Slip – better interface design
    - Mistake – better understanding of system

Abowd and Beale interaction framework



- The interaction framework involves 4 parts:

  User, Input, System, Output

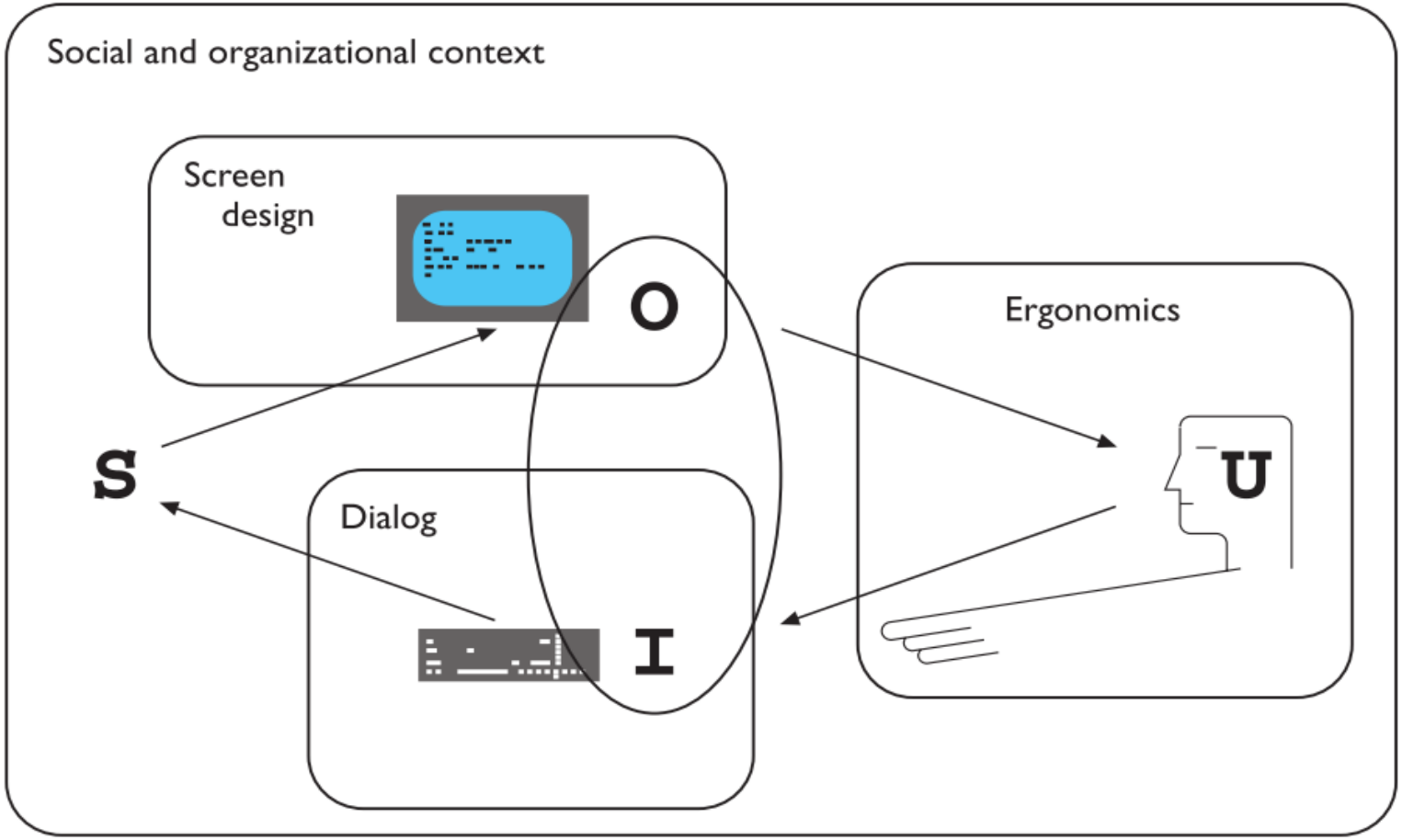
- How it works:
  - *The User begins the interactive cycle with the formulation of a goal and a task to achieve that goal*
  - *The user interacts the machine through providing inputs into it*
  - *The System then transforms itself by the operations (the execution phase) and results a new state*
  - *The current values of system attributes (output) are rendered/communicated to the user,*
    - and *then it is up to the User to observe* and assess the results of the interaction relative to the original goal, ending the evaluation phase and, hence, the interactive cycle.

Frameworks and HCI

- Discusses design issues that affects the interaction
  - *the design of interactive systems and the performance of the user*
- These issues are:
  - ***Ergonomics***
    - Addresses issues the user side of the interface, input and output and the user's immediate context
  - ***Interaction styles***
    - relates to the input branch of the framework
  - ***Presentation and screen design***
    - relates to the input branch of the framework
  - ***Social and organizational context***
    - within w/c the entire framework can be placed

Frameworks and HCI ...

- A framework for human–computer interaction



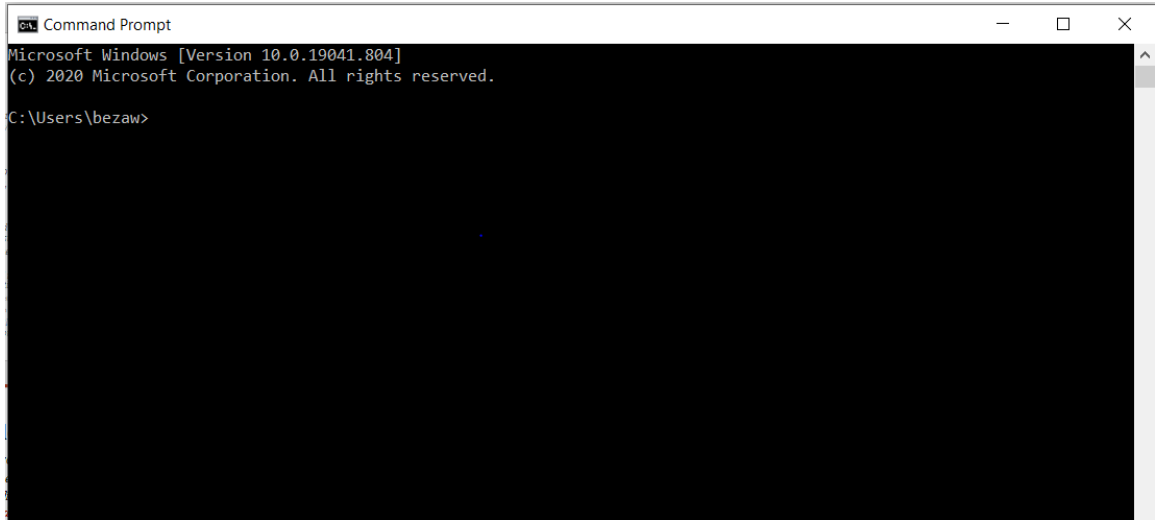
Interaction styles

- Creates a dialogue between the computer and the user
  - *Provides an interface between the user and the computer*
  - *D/t applications have distinct styles of interaction*
  - *It focuses for issue for dialogue design*
- Common interaction styles
  - *Command line interface*
  - *Menus*
  - *Natural language*
  - *Question/answer and query dialogue*
  - *Form-fills and spreadsheets*
  - *WIMP*
  - *Point and click*
  - *Three–dimensional interfaces*

Command line interface

- Way of expressing instructions to the computer directly
  - *Function keys, single characters, short abbreviations, whole words, or a combination*
- Suitable for repetitive tasks
- Better for expert users than novices (beginners)
- Offers direct access to system functionality
- Command names/abbreviations should be meaningful!

Typical example: the Unix system, DOS



Menus

- Menu driven interface: set of options displayed on the screen
- Options visible
  - *Less recall - easier to use*
  - *Rely on recognition so names should be meaningful*
- Selection by:
  - *Numbers, letters, arrow keys, mouse*
  - *Combination (e.g. mouse plus accelerators)*
- Often options hierarchically grouped
  - *Sensible grouping is needed*
- Restricted form of full WIMP system

Natural language

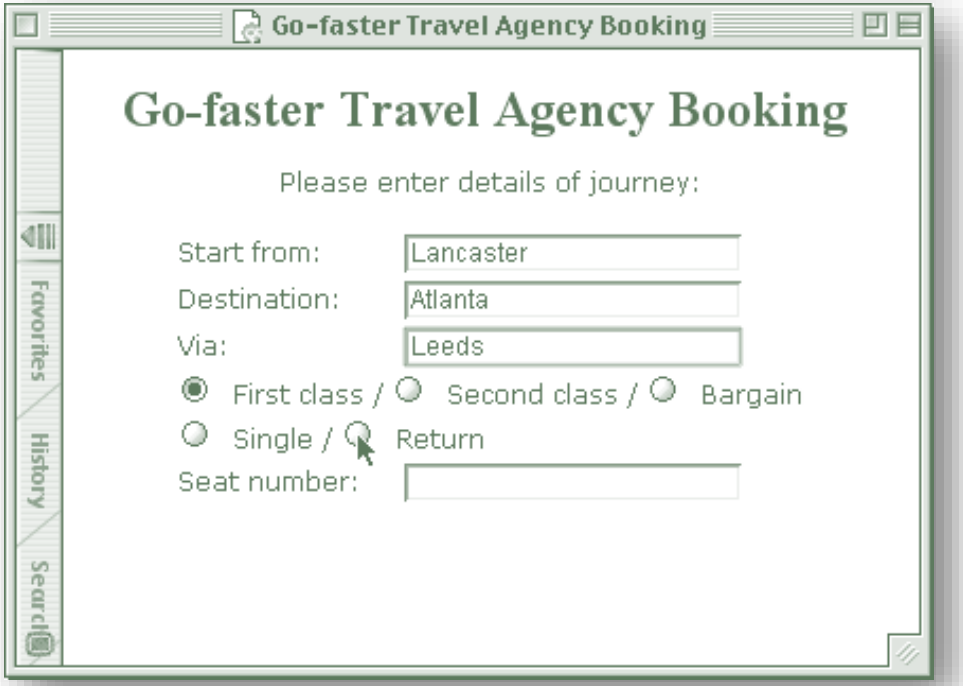
- Familiar to user- an attractive option
- Speech or typed natural language can be used as an input
- Problems
  - *Vague*
  - *Ambiguous*
  - *Hard to do well!*

Query interfaces

- Question/answer interfaces
  - *User led through interaction via series of questions*
  - *Suitable for novice users but restricted functionality*
  - *Often used in information systems*
- Query languages (e.g. SQL) are highly used

Form-filling interfaces

- Primarily for data entry or data retrieval
- Screen like paper form.
- Data put in relevant place
- Requires
  - *Good design*
  - *Obvious correction facilities*



Spreadsheets

- First spreadsheet VISICALC, followed by Lotus 1-2-3, MS Excel (most common today)
- Sophisticated variation of form-filling.
  - *Grid of cells contain a value or a formula*
  - *Formula can involve values of other cells*
    E.g. sum of all cells in this column
  - *User can enter and alter data spreadsheet maintains consistency*

Point and click interfaces

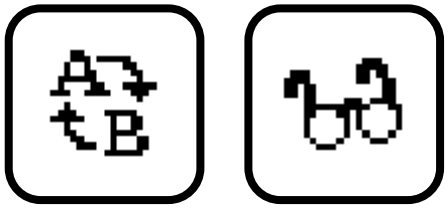
- Used in:
  - *Multimedia*
  - *Web browsers*
  - *Hypertext*

- Just point and click something!
  - *Icons, text links or location on map*

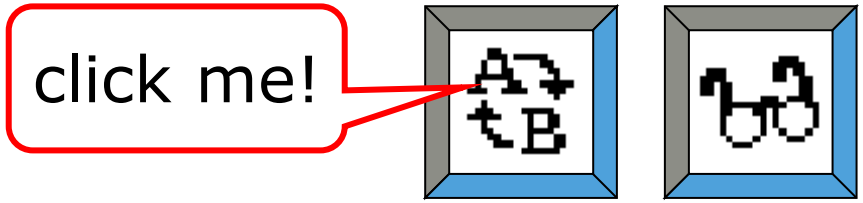
- Minimal typing

Three dimensional interfaces

- Virtual reality
- ‘Ordinary’ window systems
  - *Highlighting*
  - *Visual affordance*
- 3D workspaces
  - *Use for extra virtual space*
  - *Distance effects*



flat buttons …



… or sculptured

WIMP Interface

Windows

Icons

Menus

Pointers

… or windows, icons, mice, and pull-down menus!

- Default style for majority of interactive computer systems, especially PCs and desktop machines

ELEMENTS OF THE WIMP INTERFACE

windows, icons, menus, pointers

Buttons, toolbars,
Palettes, dialog boxes

Windows

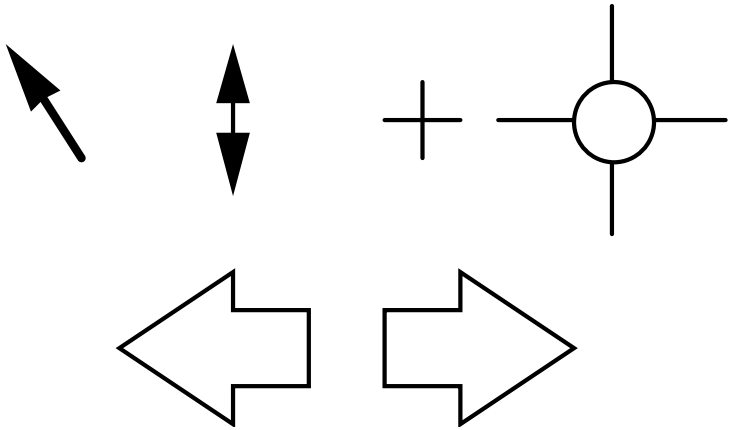
- Areas of the screen that behave as if they were independent
  - *Can contain text or graphics*
  - *Can be moved or resized*
  - *Can overlap and obscure each other, or can be laid out next to one another (tiled)*
- Scrollbars
  - *Allow the user to move the contents of the window up and down or from side to side*
- Title bars
  - *Describe the name of the window*

Icons

- Small picture or image
- Represents some object in the interface
  - *Often a window or action*
- Windows can be closed down (iconised)
  - *Small representation*
- Icons can be many and various
  - *Highly stylized*
  - *Realistic representations.*

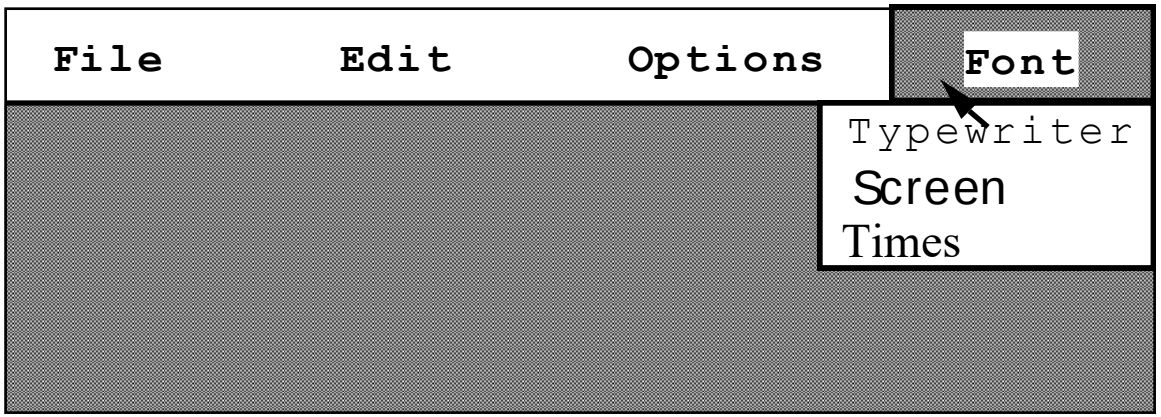
Pointers

- Important component
  - *WIMP style relies on pointing and selecting things*
- Uses mouse, trackpad, joystick, trackball, cursor keys or keyboard shortcuts
- Wide variety of graphical images



Menus

- Choice of operations or services offered on the screen
- Required option selected with pointer



Problem – take a lot of screen space

Solution – pop-up: menu appears when needed

Kinds of Menus

- Menu Bar at top of screen (normally), menu drags down
  - ***Pull-down menu*** - *mouse hold and drag down menu*
  - ***Drop-down menu*** - *mouse click reveals menu*
  - ***Fall-down menus*** - *mouse just moves over bar!*

- Contextual menu appears where you are
  - ***Pop-up menus*** - *actions for selected object*
  - ***Pie menus*** - *arranged in a circle*
    - Easier to select item (larger target area)
    - Quicker (same distance to any option)
      … but not widely used!
    - Common in smart phones

Menus extras

- Cascading menus
  - *Hierarchical menu structure*
  - *Menu selection opens new menu*
  - *and so on*

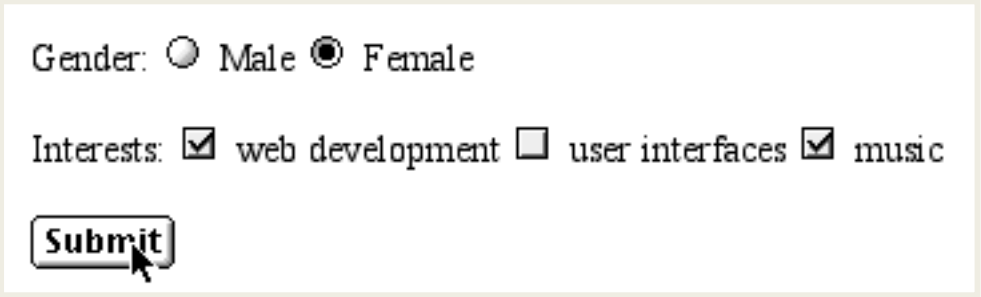
- Keyboard accelerators
  - *Key combinations - same effect as menu item*
  - *Two kinds*
    - Active when menu open – usually first letter
    - Active when menu closed – usually Ctrl + letter

Menus design issues

- Which kind to use
- What to include in menus at all
- Words to use (action or description)
- How to group items
- Choice of keyboard accelerators

Buttons

- Individual and isolated regions within a display that can be selected to invoke an action



- Special kinds
  - *Radio buttons*
    - *Set of mutually exclusive choices*
  - *Check boxes*
    - *Set of non-exclusive choices*

Toolbars

- Long lines of icons
- Fast access to common actions
- Often customizable:
  - *Choose which toolbars to see*
  - *Choose what options are on it*

Palettes-little windows of actions

- *Shown/hidden via menu option*
  E.g. available shapes in drawing package

Dialogue boxes

- Information windows that pop up to inform of an important event or request information.

E.g: when saving a file, a dialogue box is displayed to allow the user to specify the filename and location.

Once the file is saved, the box disappears.

Context of interaction

- Interaction is not JUST takes place in a vacuum
- Takes place within social and organizational context
- **Social factors**
  - *The presence of other people in a work environment affect the performance of the user/worker*
    - Develop competition-behaviour
    - Fear(frustration) drops productivity
    - Lack of motivation
- **Organizational factors**
  - *lose motivation if a system is introduced that does not match the actual requirements of the job to be done*
    - reject the system,
    - be resentful and unmotivated
    - adapt the intended interaction to his/her own requirements.
- **Solutions**: :
  - *The system must keep personal requirements, safety , etc*
  - *Make it easy for them!*
  - *Understand their values: individuals value → organisational value*

PARADIGMS OF INTERACTION

Paradigm Shifts

Paradigms of interaction

- A paradigm is a way of thinking about the word
  - It may be Predominant theoretical frameworks or scientific world views
- Occurrence of paradigm shifts
  - New perception of the human - computer relationship due to advancement of technologies
  - Changes in social, organizational, mental aspects
- We can trace some of these shifts in the history of interactive technologies
  - History will judge which are the true shifts

Why study paradigms?

- ***Goals of design***:
  - Designing for maximum usability
- Concerns/issues?
  - *How can an interactive system be developed to ensure its usability?*
  - *How can the usability of an interactive system be demonstrated or measured?*

Why study paradigms?

- To build effective strategies for designing usable interactive systems.
- To study history of interactive system design as it provides paradigms for usable designs
- To understanding HCI history since it is largely about understanding a series of paradigm shifts
  - *paradigms range from the introduction of batch processing to time sharing, to the WIMP and web, to ubiquitous and context-aware computing.*

REMARKABLE PARADIGM SHIFTS

From batch processing to Sensor-based and Context-aware Interaction

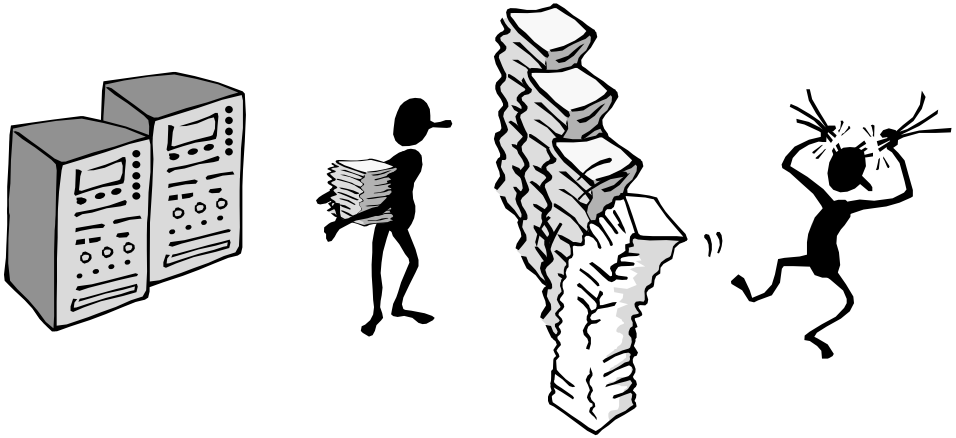
The initial paradigm

- Batch processing

-Earliest computer

-Extremely expensive devices

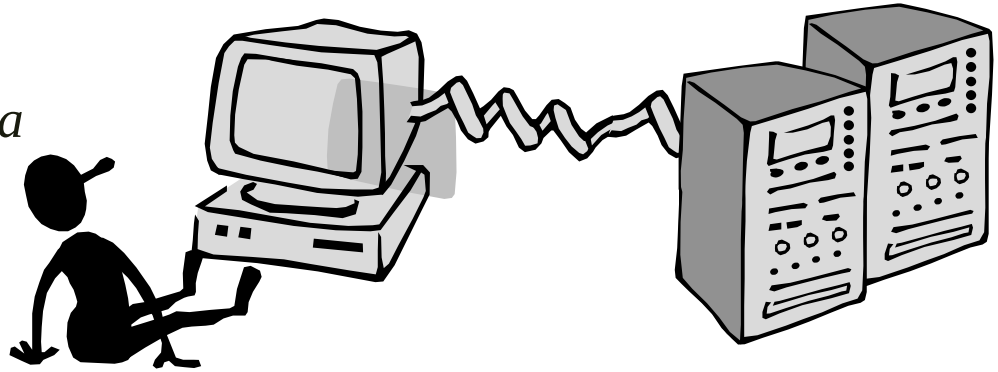
- Machine dedicatated to particular set of tasks and operated by control panel
- Complete jobs processed individually.



Impersonal computing

Example Paradigm Shifts

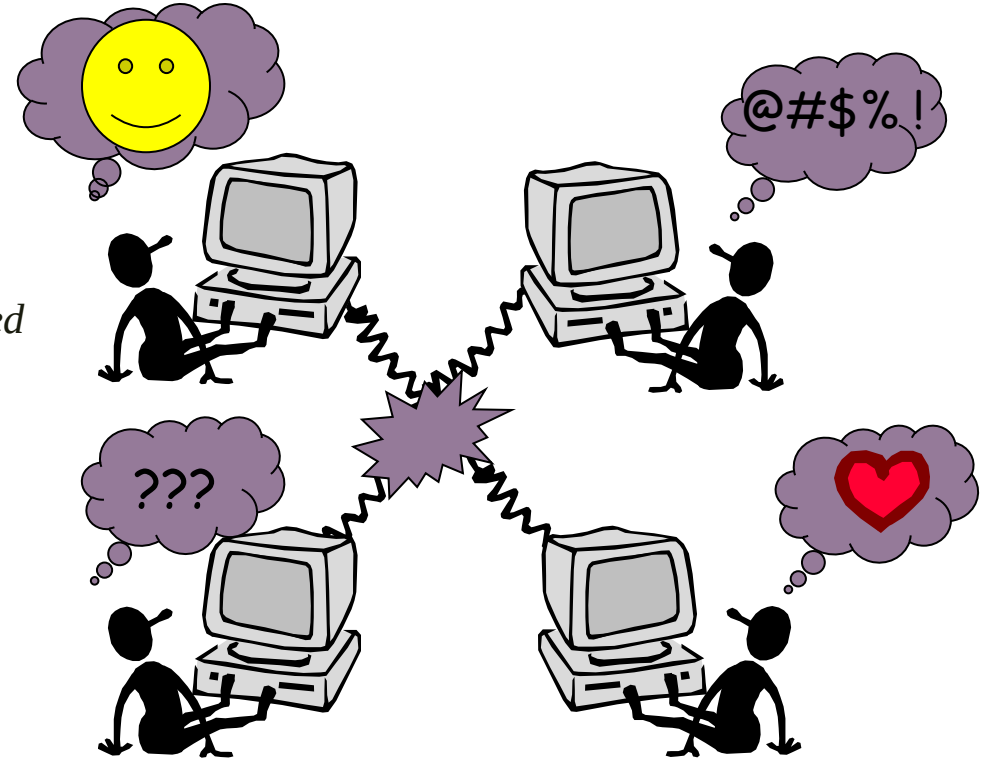
- Batch processing
- Time-sharing
  - *Many people, located at various terminals, to use a particular **computer system** at the same **time**.*
  - *Processor's **time** which is **shared** among multiple users*



Interactive computing

Example Paradigm Shifts

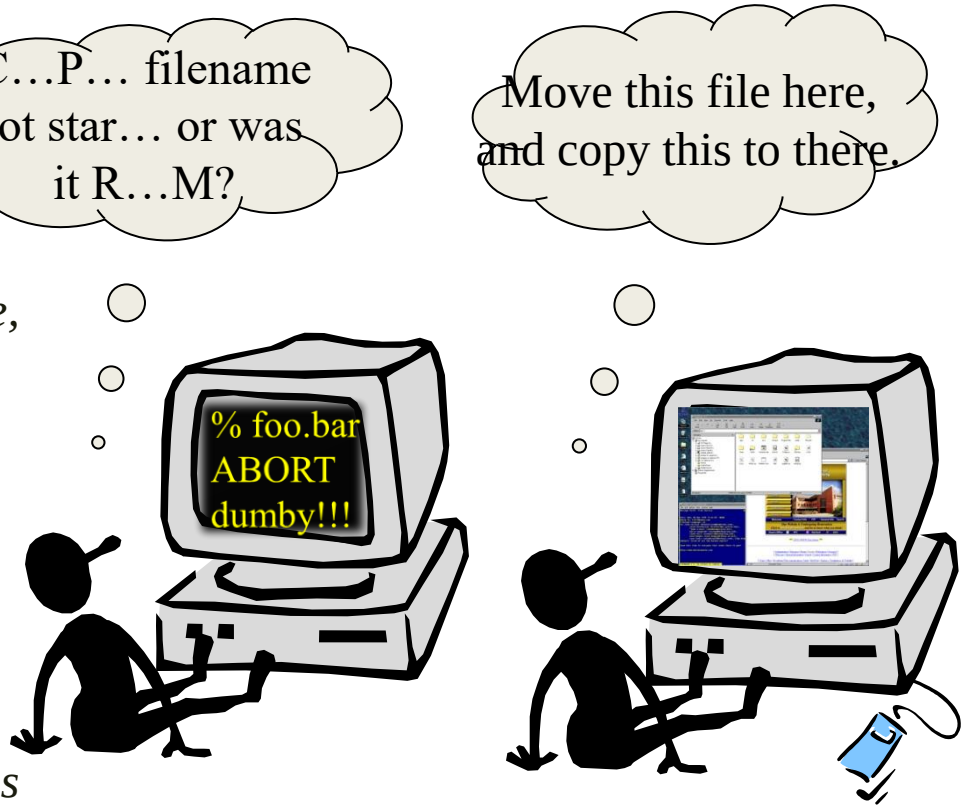
- Batch processing
- Timesharing
- Networking
  - *Social Interaction in Networked Communities*
  - *A resource-sharing model in which users are provided with free or low-cost computer and Internet access*



Community computing

Example Paradigm Shifts

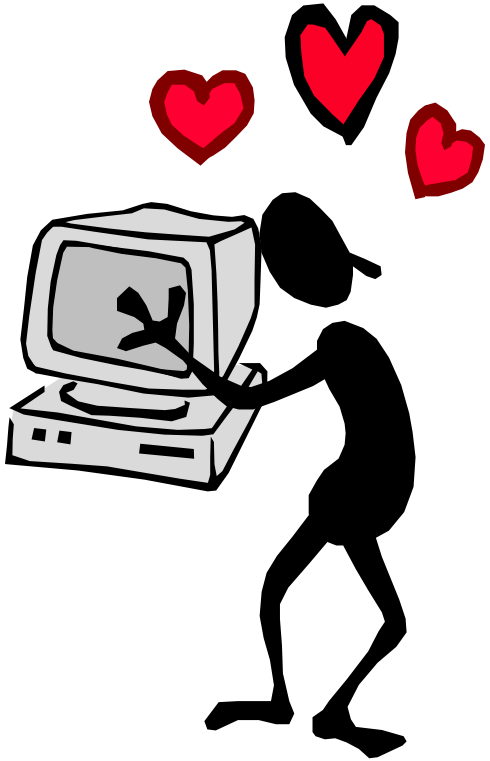
- Batch processing
- Timesharing
- Networking
- Graphical displays
  - *representation of objects of interest and rapid, reversible, and incremental actions and feedback*
  - *the user feel as if he/she physically interacts with their operating system*
  - *resizing a graphical shape, such as a rectangle, by dragging its corners or edges with a mouse*



Direct manipulation

Example Paradigm Shifts

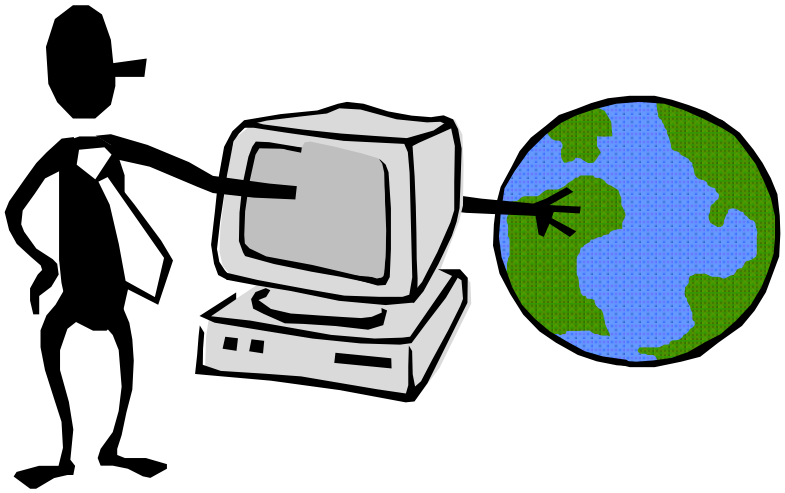
- Batch processing
- Timesharing
- Networking
- Graphical display
- Microprocessor
  - *Different types and size of work stations, handheld computers, GPS, etc with one or more operating systems that help runs multiple personal applications*



Personal computing

Example Paradigm Shifts

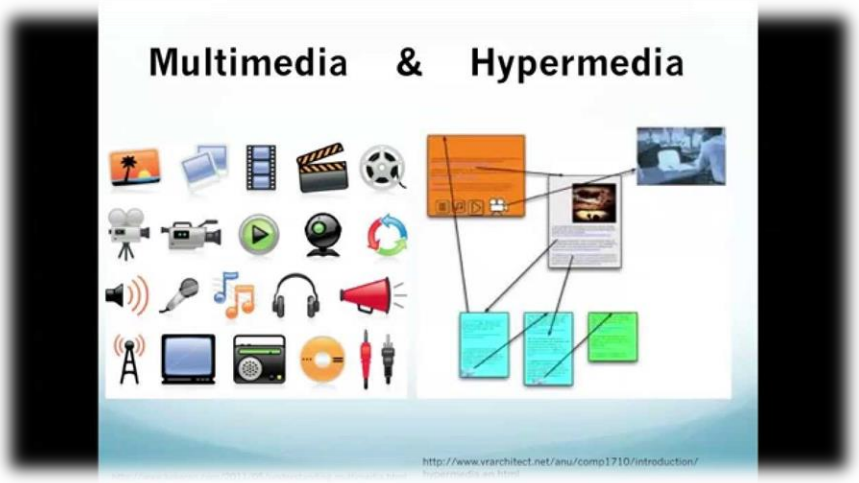
- Batch processing
- Timesharing
- Networking
- Graphical display
- Microprocessor
- WWW
  - *Interact to the world using* ***internet***

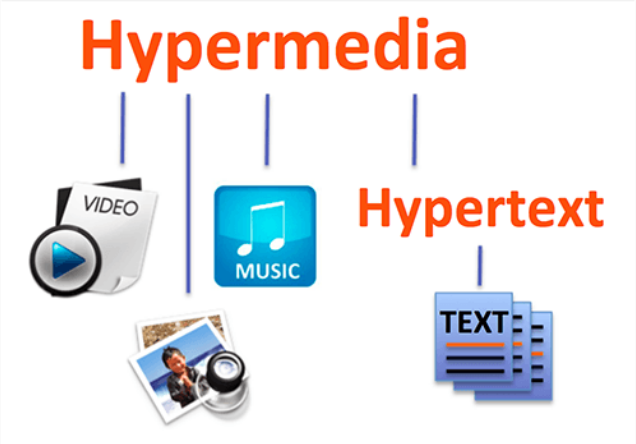


Global information

Hypertext

- key to success in managing explosion of information
- non-linear browsing structure
- hypermedia and multimedia



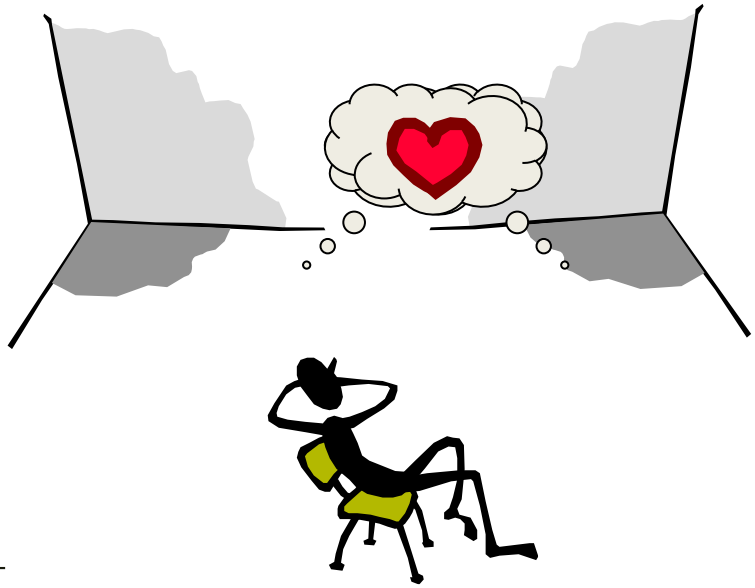


The World Wide Web/Internet

- Hypertext, as originally realized, was a closed system
- Simple, universal protocols (e.g. HTTP) and mark-up languages (e.g. HTML) made publishing and accessing easy
- Critical mass of users lead to a complete transformation of our information economy.

Example Paradigm Shifts

- Batch processing
- Timesharing
- Networking
- Graphical display
- Microprocessor
- WWW
- Ubiquitous Computing
  - A interaction of **physical** and **electronic worlds** in service of everyday activities.



More on: http://www.rheingold.com/texts/tft

Multimodality

- Use of multiple language for interaction (virtual, physiscal)
- Simultaneous use of multiple channels for input and output
  - *Textual*
  - *Aural*
  - *Linguistic*
  - *Visual*
  - *Hand writing*
  - *Hand gesture*
  - *Gaze*
  - *etc*
- Designing a multimodal user interface is a research area in HCI

Others

- Computer Supported Cooperative Work (CSCW)
- Agent-based Interfaces
- Ubiquitous Computing
- Sensor-based and Context-aware Interaction

THANK YOU
THANK YOU!
THANK YOU